Compressed Sensing of Sparsity-constrained Total Variation Minimization for CT Image Reconstruction

Jian Dong[a], Hiroyuki Kudo[a, b], Essam A. Rashed[c]

[a] Faculty of Engineering, Information and Systems, University of Tsukuba, Tennoudai 1-1-1, Tsukuba 305-8573, Japan

[b] JST-ERATO Momose Quantum-Beam Phase Imaging Project, Katahira 2-1-1, Aoba-ku, Sendai 980-8577, Japan

[c] Image Science Lab., Department of Mathematics, Faculty of Science, Suez Canal University, Ismailia 41522, Egypt

ABSTRACT

Sparse-view CT image reconstruction is becoming a potential strategy for radiation dose reduction of CT scans. Compressed sensing (CS) has been utilized to address this problem. Total Variation (TV) minimization, a method which can reduce streak artifacts and preserve object boundaries well, is treated as the most standard approach of CS. However, TV minimization cannot be solved by using classical differentiable optimization techniques such as the gradient method, because the expression of TV (TV norm) is non-differentiable. In early stages, approximated solving methods were proposed by changing TV norm to be differentiable in the way of adding a small constant in TV norm to enable the usage of gradient methods. But this reduces the power of TV in preserving accuracy object boundaries. Subsequently, approaches which can optimize TV norm exactly were proposed based on the convex optimization theory, such as generalizations of the iterative soft-thresholding (GIST) algorithm and Chambolle-Pock algorithm. However, these methods are simultaneous-iterative-type algorithms. It means that their convergence is rather slower compared with row-action-type algorithms. The proposed method, called sparsity-constrained total variation (SCTV), is developed by using the alternating direction method of multipliers (ADMM). On the method we succeeded in solving the main optimization problem by iteratively splitting the problem into processes of row-action-type algebraic reconstruction technique (ART) procedure and TV minimization procedure which can be processed using Chambolle's projection algorithm. Experimental results show that the convergence speed of the proposed method is much faster than the conventional simultaneous iterative methods.

Keywords: Image Reconstruction, Computed Tomography, Total Variation, Sparsity, ADMM

1. INTRODUCTION

This paper deals with computed tomography (CT) image reconstruction problem from sparse-view projection data. This topic has been attracting much concerns recently because it may contribute to reduce X-ray radiation dose of CT scan examinations on a large scale [1, 2, 3]. However, the incompleteness of projection data induced by sparse sampling always makes failures of the image reconstruction, such as occurrence of heavy streak artifacts or loss of object boundaries and image textures. To address the problem, a principle called compressed sensing (CS) has been proposed by Donoho [4] and Candes [5]. And to our delight, it has shown the potentiality of perfectly solving the problem. The most widely researched case of CS, which is called total variation (TV), was proposed by Rudin, Osher and Fatemi [6]. And because of its effectiveness in reducing streak artifacts and preserving accurate object boundaries, TV has become the standard approach of CS in image reconstruction field [7, 8, 9]. However, TV minimization cannot be solved by using classical differentiable optimization techniques such as the gradient method, since the expression of TV (TV norm) is non-differentiable. In early stages, approximated solving methods were proposed by changing TV norm to be differentiable in the way of adding a small constant to TV norm to enable the usage of gradient methods [10]. But this rescue measures reduced the power of TV in preserving accurate object boundaries. Subsequently, approaches which can

Further author information: (Send correspondence to Jian Dong)

: E-mail: dongjian2007happy@yahoo.co.jp

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optimize TV norm exactly were proposed based on the convex optimization theory, such as generalizations of the iterative soft-thresholding (GIST) algorithm [11], Chambolle-Pock algorithm [12] and alternating extragradient (AE) algorithm [13]. However, these methods are simultaneous-iterative-type algorithms. It means that their convergence is rather slower compared with that for row-action-type algorithms such as algebraic reconstruction technique (ART) method. Thus, demand has been growing for developing an exact TV optimization algorithm with fast convergence.

In the literature, an efficient algorithm dealing with sparsity-constrained $L1$ norm (expression like $\|\vec{x}\|_1^1 = \sum_{t=1}^{T}|x_t|$) minimization reconstruction was proposed by Rashed et al [14]. This algorithm was achieved by using the alternating direction method of multipliers (ADMM). It is highly evaluated for its ingenious structure consisting of two main building blocks. One block is ART iterative process, an iterative restoration which has attractive features such as simplicity of implementation and storage efficiency, and the other block is soft-thresholding process. Accordingly, the algorithm has a very fast convergence speed. We also appreciate the ingenious structure and fast convergence of Rashed's algorithm so much. However, the method proposed by Rashed can only reconstruct a sparsity image (majority of the image components are zeros or approximately zeros) which restricts its range of application. In this paper, we employ the ADMM method to optimize sparsity-constrained TV norm. On one hand, it broadens the application for dealing with a general image. On the other hand, the two benefits of Rashed's algorithm can be expressed efficiently.

The purpose of this paper is to propose a new algorithm of sparsity-constrained total variation (SCTV) for sparse-view image reconstruction. Similarly to the method Rashed proposed in [14], we also used the ADMM. Instead of reconstructing sparse images, our new proposal enables the application of reconstructing a general image. A trial on exactly minimizing TV norm with use of ADMM is an original contribution of this paper, as it has not been investigated yet. Numerical phantom and real CT data were used to verify the efficiency of the proposed algorithm.

2. METHODOLOGY

2.1 Problem formulation

In this section, we formulate image reconstruction problem in sparse-view CT. We denote an image by a J-dimensional vector $\vec{x} = (x_1,\ x_2,\ \cdots,\ x_J)^T$ and denote a corresponding projection data (sinogram) by an I-dimensional vector $\vec{b} = (b_1,\ b_2,\ \cdots,\ b_I)^T$. We denote the system matrix which relates $\vec{x}$ to $\vec{b}$ by $\mathrm{A} = \{a_{ij}\}$. The system matrix A is an $I \times J$ matrix. When the dimension of $\vec{b}$ is smaller than the dimension of $\vec{x}$ (*i.e.* $I < J$), the problem becomes underdetermined and reconstructing an accurate image becomes challenging. This is just the mathematical setup of sparse-view CT reconstruction. In this situation, CS usually works for obtaining a feasible image by minimizing the following linearly constrained cost function:

$$\min_{\vec{x}\in\mathrm{R}^J} f(\vec{x}) \qquad \text{subject to } A\vec{x} = \vec{b} \tag{1}$$

where $f(\vec{x})$ is the cost function to pick up a reasonable image $\vec{x}$ with less streak artifacts from a set of candidate solutions satisfying $A\vec{x} = \vec{b}$. For this purpose, many cost functions have been proposed in the literature such as Gibbs prior [15], median root prior [16], TV [8, 9], and image sparsity prior [17]. The cost functions of Gibbs prior and median root prior have been disclosed to have defects in dealing with the task of image reconstruction. Since they carry out the same level smoothing on the whole image, which leads to a failure in preserving accurate object boundaries. In the literature [17], Li proposed the image sparsity prior which can be expressed in Eq. (2).

$$f(\vec{x}) = \|\vec{x}\|_1^1 \equiv \sum_{j=1}^{J}|x_j| \tag{2}$$

In this prior, the cost function is called $L1$ norm of $\vec{x}$. It is well known that the value of Eq. (2) becomes small when $\vec{x}$ is sparse and its power to find a sparse solution is surprisingly strong in contrast to the ordinary $L2$ norm of $\vec{x}$ (*i.e.* $\|\vec{x}\|_2^2 \equiv \sum_{j=1}^{J}|x_j|^2$). Rashed also used this cost function in developing a leading method of three-dimensional image reconstruction for C-arm angiography [14]. Moreover, Rashed obtained the iterative algorithm by using the ADMM principle which we appreciate because of its fast convergence and simplicity in implementation. However, this image sparsity prior cost function can take effect only in reconstructing a sparsity image. But as we know, most of medical images are not sparse, which restricts its range of application. On the other hand, the cost function of TV has become the gold standard in CS filed in the task of image reconstruction. It is effective in removing streak artifacts and preserving object boundaries. Accordingly, we deal with the following TV cost function in this paper.

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$$f(\vec{x}) = \|\vec{x}\|_{TV} = \sum_m \sum_n \sqrt{(x_{m,n} - x_{m+1,n})^2 + (x_{m,n} - x_{m,n+1})^2} \tag{3}$$

We regard the image as an $N \times N$ pixel image in the definition of Eq. (3). $x_{m,n}$ denotes the pixel value of image $\vec{x}$ at pixel coordinate (m,n). It is easy to find that the TV norm is a non-differentiable expression. And considerable efforts are needed to minimize the TV norm strictly. Inspired by the process mode in literature [14], in this paper, we propose and implement an alternating iterative algorithm derived from the optimization method ADMM which is attracting a lot of attention recently. In this mode, the popular ART technic can be used to guarantee the fast convergence.

2.2 TV minimization via ADMM

In this section, we explain how to solve the optimization problem of Eq. (1) with Eq. (3) by means of ADMM procedure. A nice review of ADMM can be found in literature [14], and please refer to it for a good understanding of the derivation in this section.

We can redefine the problem by introducing a new variable $\vec{z}$, and the new problem definition becomes as follows:

$$\min_{(\vec{x},\vec{z})} f(\vec{x}) + g(\vec{z}) \qquad subject\ to\ \vec{x}\text{-}\vec{z}=\vec{0} \tag{4}$$

where $f(\vec{x})$ and $g(\vec{z})$ are defined as:

$$f(\vec{x}) = \begin{cases} 0 & if\ A\vec{x} = \vec{b} \\ \infty & otherwise \end{cases} \qquad \text{and} \qquad g(\vec{z}) = \|\vec{z}\|_{TV} \tag{5}$$

We note that $f(\vec{x})$ is the characteristic function corresponding to the affine subspace $A\vec{x} = \vec{b}$. It can be easily confirmed that the solution to Eq. (4) with Eq. (5) coincides with the solution to Eq. (1) with Eq. (3). In ADMM, the iterative algorithm is constructed in order to solve this problem according to the following procedure. First, we define the augmented Lagrangian function corresponding to the problem of Eq. (4) by

$$L_\rho(\vec{x},\vec{z},\vec{\lambda}) = f(\vec{x}) + g(\vec{z}) + \vec{\lambda}^T(\vec{x}\text{-}\vec{z}) + \frac{\rho}{2}\|\vec{x} - \vec{z}\|_2^2 \tag{6}$$

where $\vec{\lambda}$ is a *J*-dimensional vector called the dual vector or the Lagrange multiplier vector and ρ>0 is the weighting parameter of the quadratic penalty term. Using Eq. (6), the iterative algorithm can be expressed as the following three-step procedure (with *k* as the main iteration number):

$$\begin{cases} \vec{x}^{(k+1)} = \underset{\vec{x}}{\operatorname{argmin}}\, L_\rho(\vec{x},\vec{z}^{(k)},\vec{\lambda}^{(k)}) = \underset{\vec{x}}{\operatorname{argmin}} \left[f(\vec{x}) + \vec{\lambda}^{(k)^T}(\vec{x} - \vec{z}^{(k)}) + \frac{\rho}{2}\|\vec{x} - \vec{z}^{(k)}\|_2^2\right] & (7) \\ \vec{z}^{(k+1)} = \underset{\vec{z}}{\operatorname{argmin}}\, L_\rho(\vec{x}^{(k+1)},\vec{z},\vec{\lambda}^{(k)}) = \underset{\vec{z}}{\operatorname{argmin}} \left[g(\vec{z}) + \vec{\lambda}^{(k)^T}(\vec{x}^{(k+1)} - \vec{z}) + \frac{\rho}{2}\|\vec{x}^{(k+1)} - \vec{z}\|_2^2\right] & (8) \\ \vec{\lambda}^{(k+1)} = \vec{\lambda}^{(k)} + \rho(\vec{x}^{(k+1)} - \vec{z}^{(k+1)}) & (9) \end{cases}$$

As the attractive feature of this algorithm, the minimization of $f(\vec{x}) + g(\vec{z})$ can be split into the sequence of two separate minimizations with respect to $f(\vec{x})$ and $g(\vec{z})$. This is a powerful framework of optimization, especially in this study where $f(\vec{x})$ is a characteristic function and $g(\vec{z})$ is TV norm, both of which are not strongly convex.

The algorithm expressed by Eqs. (7) - (9) still does not have an explicit implementable form. So we further simplify them below. First, since $f(\vec{x})$ is the characteristic function corresponding to $A\vec{x} = \vec{b}$, the solution $\vec{x}^{(k+1)}$ to Eq. (7) can be achieved by solving the following linearly constrained optimization:

$$\min_{\vec{x}} \left\|\vec{x} - (\vec{z}^{(k)} - \vec{\lambda}^{(k)}/\rho)\right\|_2^2 \text{ s.t. to } A\vec{x} = \vec{b} \tag{10}$$

Since Eq. (10) is the standard Euclidean distance minimization subject to a linear equality constrained, a variety of methods in the linear algebra field can be used to solve it. However, the problem size is so large that we need to adopt a storage efficient method. In this work, we use ART iterative method with the initial vector $\vec{z}^{(k)} - \vec{\lambda}^{(k)}/\rho$, which is known to be efficient and easy to implement. The iterative formula can be shown as below:

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$$\vec{x}^* = \mathrm{ART}\left(\vec{z}^{(k)} - \vec{\lambda}^{(k)}/\rho\right) \tag{11}$$

The detailed algorithm for the $\vec{x}$ minimization with use of ART can be summarized in **Algorithm I**.

Algorithm I. ART method to solve the sub-problem of Eq. (10)

Step 1. Initialization: $\vec{y}^{(0)} = \vec{z}^{(k)} - \vec{\lambda}^{(k)}/\rho$
Step 2. Mediate image $\vec{y}$ Update
for n=0, 1, $\cdots$, $N-1$ **do**
$\vec{y}^{(n,0)} = \vec{y}^{(n)}$
for i=0, 1, $\cdots$, $I-1$ **do**
$\vec{y}^{(n,i+1)} = \vec{y}^{(n,i)} + \frac{b_i - \vec{a}_i^T \vec{y}^{(n,i)}}{\|\vec{a}_i\|_2^2} \vec{a}_i^T$
end for
$\vec{y}^{(n+1)} = \vec{y}^{(n,I)}$
end for
Step 3. Solution $\vec{x}^* = \vec{y}^{(N)}$

It is known that ART can be implemented with a very fast convergence rate by using some special data access order [18]. This behavior is due to the increase of the orthogonality among projections at each iteration and is confirmed by several later studies [19, 20]. We use this data access order in this study.

Next, we describe how to solve the $\vec{z}$ minimization of Eq. (8). We can first equivalently convert the minimization problem of Eq. (8) as below:

$$\min_{\vec{z}} \ \|\vec{z}\|_{TV} + \frac{\rho}{2}\left\|\vec{z} - (\vec{x}^{(k+1)} + \vec{\lambda}^{(k)}/\rho)\right\|_2^2 \tag{12}$$

The expression in Eq. (12) is the sum of TV norm and a quadratic term, and it frequently appears in the CS literature and be regarded as the TV denoising problem with a degraded image vector $(\vec{x}^{(k+1)} + \vec{\lambda}^{(k)}/\rho)$. The $\vec{z}$ minimization in Eq. (12) can be solved compactly by a standard algorithm of this area called Chambolle's projection algorithm with the initial vector $\vec{x}^{(k+1)} + \vec{\lambda}^{(k)}/\rho$. On the common sense, Chambolle's projection algorithm is generally rated as an accurate and efficient procedure to solve the TV denoising problem [21, 22]. The solution can be summarized by

$$\vec{z}^* = \mathrm{TV}(\vec{x}^{(k+1)} + \vec{\lambda}^{(k)}/\rho) \tag{13}$$

In our implementations, we used Chambolle's projection algorithm. Following the image update of $\vec{z}$, the Lagrange multiplier $\vec{\lambda}$ is updated as Eq. (9).

Basically, the proposed algorithm is given by Eqs. (7) - (9), and we showed that the $\vec{x}$ minimization in Eq. (7) can be performed using the ART technic and the $\vec{z}$ minimization can be degraded to TV denosing problem which can be compactly solved by Chambolle's projection algorithm. Therefore, the final algorithm can be summarized as **Algorithm II**.

The proposed SCTV algorithm possesses a favorable structure in the sense that its major operation consists of the ART iteration and the TV denoising iteration which can be implemented using the standard algorithm called Chambolle's projection algorithm. Both of the two building blocks (ART iteration and TV denoising iteration) are very popular in recent research of tomography image reconstruction area. On the other hand, the main drawback of SCTV is that it has a double-loop (second-order) structure so that the convergence to the true solution can only be guaranteed when the inner iterations are terminated after enough iteration numbers.

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Algorithm II. The proposed algorithm SCTV

Step 1. Initialization: $\vec{x}^{(0)} = \vec{z}^{(0)} = (0, 0, \dots, 0)^T, \vec{\lambda}^{(0)} = (0, 0, \dots, 0)^T$
Sept 2. Alternatively implementing of image update
for k=0, 1, …, $k_{max} - 1$ **do**
Update $\vec{x}$: $\vec{x}^{(k+1)} = \mathrm{ART}\left(\vec{z}^{(k)} - \vec{\lambda}^{(k)}/\rho\right)$, *i.e.* implementation of **Algorithm I**.
Update $\vec{z}$: $\vec{z}^{(k+1)}$ =TV $(\vec{x}^{(k+1)} + \vec{\lambda}^{(k)}/\rho)$, *i.e.* implementation of Chambolle's projection algorithm.
Update $\vec{\lambda}$: $\vec{\lambda}^{(k+1)} = \vec{\lambda}^{(k)} + \rho(\vec{x}^{(k+1)} - z^{(k+1)})$
end for
Step 3. Solution $\vec{z}^{*}$=$\vec{z}^{(k_{max})}$

3. EXPERIMENTAL RESULTS

3.1 Simulation with numerical phantom

We carried out simulation studies using a numerical phantom called "spot phantom". This numerical phantom contains circles and rectangles with various contrast and sizes. It was designed for the investigation of our new SCTV algorithm which is used in sparse-view CT image reconstruction. The image size was 256×256 pixels and the projection data was acquired by parallel-beam geometry with 256 radial bins and 32 views over 180 angular range. No statistical noise was added to the projection data in the simulations. In this study, we calculated the Root Mean Squares Error (RMSE) value as the quantitative evaluating metric. This metric indicates the similarity between the reconstructed image and the true image. The formula of this metric is defined as in Eq. (14).

$$\mathrm{RMSE} = \sqrt{\frac{\sum_{j=1}^{J}(x_j - x_j^{(true)})^2}{J}} \tag{14}$$

where x_j denotes the reconstructed pixel value and $x_j^{(true)}$ denotes the true pixel value. A smaller resulting value indicates a better reconstruction result.

We considered two scenarios to test the efficiency of the proposed SCTV algorithm and they are summarized as follows.

[Scenario 1] We investigated the differences in image convergence speed among different ρ which denotes the weighting parameter of the quadratic penalty. A prerequisite was set that the inner iterations of ART and TV denoising procedures were implemented sufficiently. 32 projection views were used for the reconstruction. The results are shown in **Fig. 1**. RMSE values are shown on the top left corner of each result image. We observed that, under the condition of guaranteeing the inner iterations be implemented sufficiently, only a few main iterations were necessary for image convergence. When the parameter ρ was set too small, more iterations were needed, *i.e.* the convergence speed was slow. And the convergence speed was accelerated as the parameter ρ was set to a larger value. Furthermore, when ρ was set too large, the convergence speed became slow again. Thus, it is necessary to determine an appropriate value for parameter ρ to make the algorithm be of high-performance.

[Scenario 2] We compared the convergence speed of the conventional generalizations of the iterative soft-thresholding (GIST) algorithm and the proposed SCTV algorithm. In this simulation, with the purpose of saving the computational time, we took means of terminating the inner ART iterations and TV denoisnig iterations early for the SCTV algorithm. In detail, we implemented 10 inner iterations for ART and 50 iterations for TV denoising in the first two main iterations, and reduced the inner iteration of ART to only once for later main iterations. We reduced the inner ART iterations number because it is the main time-costing building block in the programing code. Here, we set $\rho = 20$ because it performed the best with $\rho = 20$ in scenario 1. The same 32 views were used for reconstruction. The results are shown

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in **Fig.2**. We can observe that the proposed algorithm only need 10 iterations for convergence, while GIST need several hundred iterations.

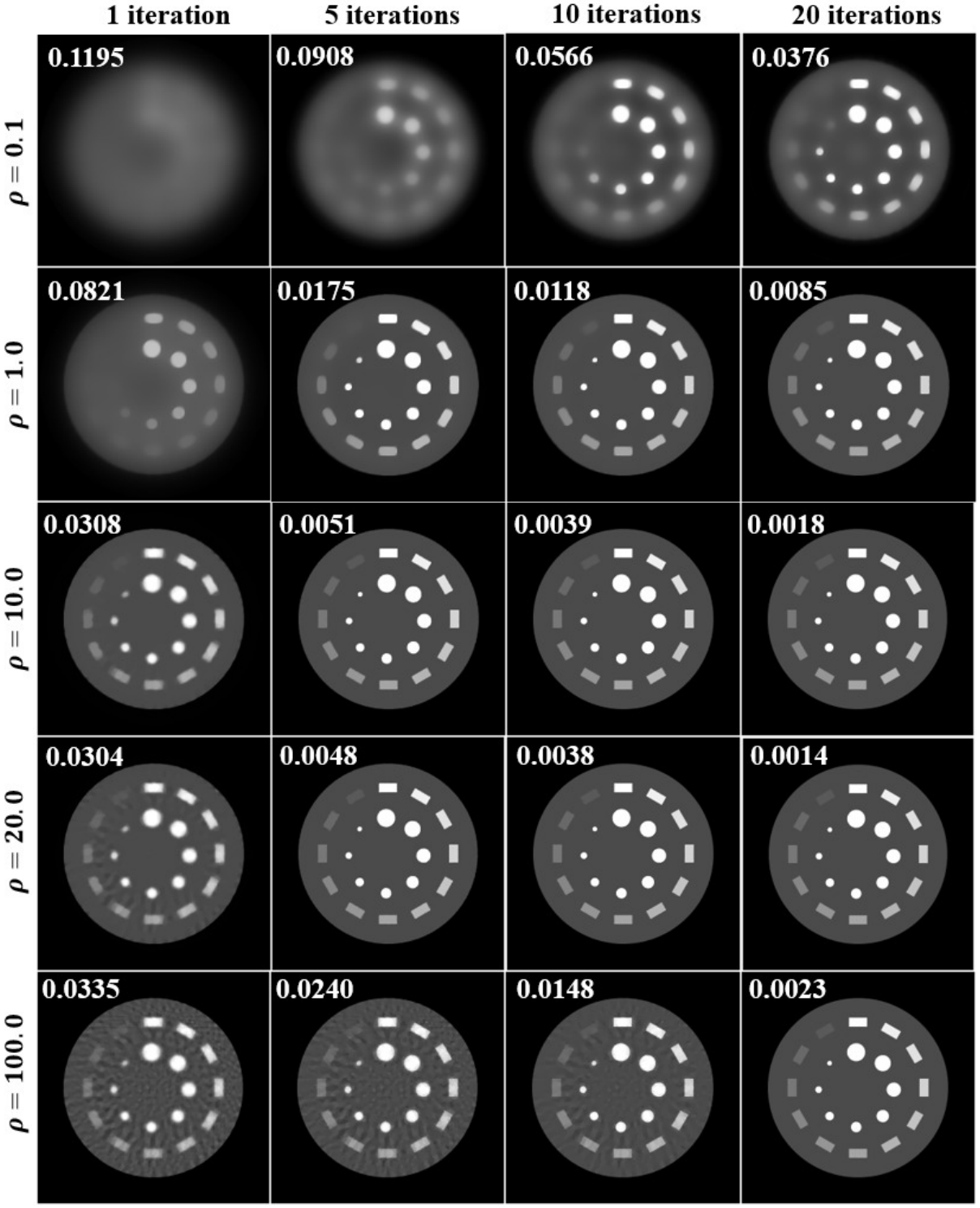

Fig.1. SCTV reconstructed images with different main iterations and ρ combinations. In each instance, the inner ART iteration and TV denoising were guaranteed to implemented sufficiently.

3.2 Simulation with practical CT image

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In this study, a slice of dental CT image was used to evaluate the convergence speed of the proposed SCTV algorithm. The image size was 512×512 and projection data was acquired by parallel-beam geometry with 512 radial bins and 48 views over 180 angular range. Similarly to the comparison of scenario 2 in section 3.1, we compared the convergence speed of the conventional GIST algorithm and the proposed SCTV algorithm. Also, with the purpose of saving the computational time, we took means of terminating the inner ART iterations and TV denoisnig iterations early for the SCTV algorithm. In detail, we implemented 10 inner iterations for ART and 50 iterations for TV denoising for the first two main iterations, and reduced the inner iteration of ART to only once for later main iterations. We set $\rho = 20$ in this experiment. The results are shown in **Fig. 3**. The high convergence speed of the proposed algorithm was also verified by the practical CT image. Similarly to the numerical phantom simulation, the practical CT image experiment also showed the trend that the new algorithm acquired only a few iterations for convergence, while several hundred iterations were needed by the conventional GIST algorithm.

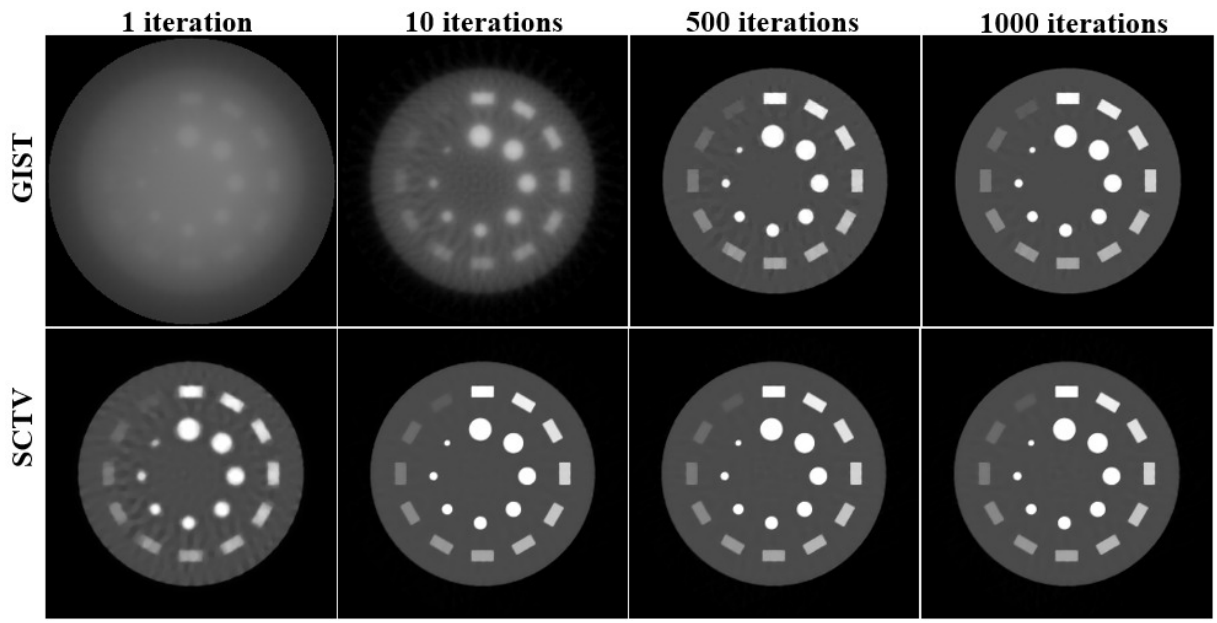

Fig.2. Reconstruction images of the conventional generalizations of the iterative soft-thresholding (GIST) algorithm and the proposed sparsity-constrained TV (SCTV) at 1, 10, 500 and 1000 iterations.

4. CONCLUSIONS

In this paper, we proposed a new image reconstruction algorithm called the sparsity-constrained total variation (SCTV) for sparse-view CT. We succeeded in achieving an exact reconstruction by SCTV. We constructed the new algorithm based on the principle called alternating direction method of multipliers (ADMM) which is attracting much attention. Thanks to the ADMM framework, we are successful in splitting the main image reconstruction task into two building blocks in each iteration, which are the conventionally famous algebraic reconstruction technique (ART) procedure and TV minimization by Chambolle's projection algorithm. The ART method is a row-action-type iterative reconstruction method which is known to have properties such as high storage efficiency and simplicity of implementation. Additionally, Chambolle's projection algorithm can reach the goal of TV minimization with a simple code. Therefore, the proposed algorithm can achieve an exact reconstruction with a high convergence rate. The effect of the new algorithm has been confirmed by experimental studies with use of both numerical phantom and practical CT image.

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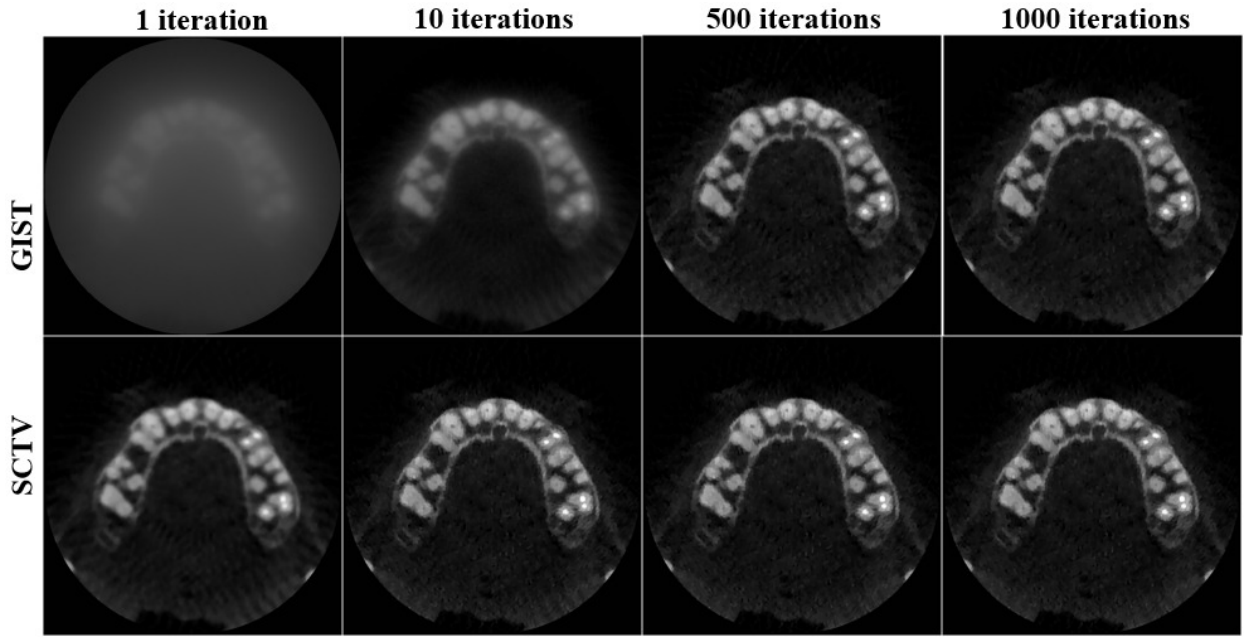

Fig.3. Reconstruction results of the practical dental CT slice by the conventional generalizations of the iterative soft-thresholding (GIST) algorithm and the proposed sparsity-constrained TV (SCTV) at 1, 10, 500 and 1000 iterations.

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